LAB 02

PH142 Fall 2026

Announcements

- **Lab02:** due Sat 9/5 at 11:59 PM (noon)
- **Quiz01:** due Fri 9/4 at 11:59 PM (midnight)
  - The quiz is available from **11:59 PM Thursday** to **11:59 PM Friday**
  - You have 30 minutes to complete the quiz

Week 2 Lecture Review

Exploring Data

- Get to know your data before analyzing!
- Key functions:
  - head(data): view the first few rows
  - dim(data): view the number of rows and columns
  - names(data): view column names
  - str(data): view the types of variables

Week 2 Lecture Review

dplyr Review (P1)

Function	Purpose	Input	Output	Notes
rename( )	Rename variables	rename(old_dataset, new_name = old_name)	Renamed dataset	Can do multiple variables at one!
select( )	Select a subset of variables	select(dataset, column1, column2)	Dataset with specific columns	Can remove variables with -
arrange( )	Sort by a variable	arrange(dataset, column)	Dataset arranged by specific columns	Can sort by multiple variables, default is ascending order, descending order with -

Week 2 Lecture Review

dplyr Review (P2)

<table>
<tr><th>Function</th><th>Purpose</th><th>Input</th><th>Output</th><th>Notes</th></tr>
<tr><td>filter()</td><td>Select a subset of rows</td><td>filter(dataset, condition)</td><td>Dataset with rows based on conditions</td><td>==, <, >, <=, >=, !=
Or (|), and(, &)</td></tr>
<tr><td>mutate()</td><td>Add new variables to a dataset</td><td>mutate(dataset, new_column = data)</td><td>Dataset with new variable</td><td>You can call existing variables in the dataset to create your new variable</td></tr>
<tr><td>group_by</td><td>Group data by a categorical variable</td><td>group_by(column)</td><td>Dataset grouped by variable</td><td>Use with summarize()!</td></tr>
<tr><td>summarize()</td><td>Summarize a statistic</td><td>summarize(statistic_name = calculation)</td><td>Tibble with grouped data and statistic for each group</td><td>Use with group_by()!</td></tr>
</table>

Week 2 Lecture Review

Pipe Operators

Key operator: Pipe (%>%)

- Connects functions together so you can do more than one in a single code chunk
- Shorthands your code by only calling the dataset once
- Read as “and then”… R will read your functions from left to right

Template: dataset %>% function1( ) %>% function2( )…

Week 2 Lecture Review

ggplot

Template: ggplot(data= dataset, aes(x=var1, y=var2)) + geom_point( )

- geom_line( )
- geom_histogram( )
- geom_bar( )

Other Changes:

- + labs(title=" " , y= " ", x=" ")
- aes(col= variable1, lty=variable2)

Week 2 Lecture Review

Types of Variables

Categorical variables: variables that have grouping levels

- Nominal variables: have no underlying order or rank
- Ordinal variables: can be ordered or ranked

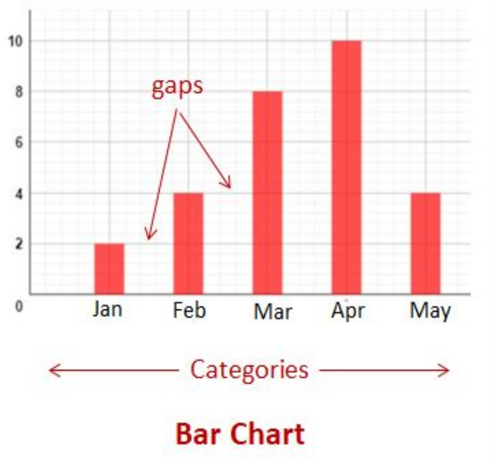
Quantitative variable: continuous, numeric variables that that you can perform mathematical operations on

- Discrete variables: can be counted
- Continuous variables: can be measured precisely

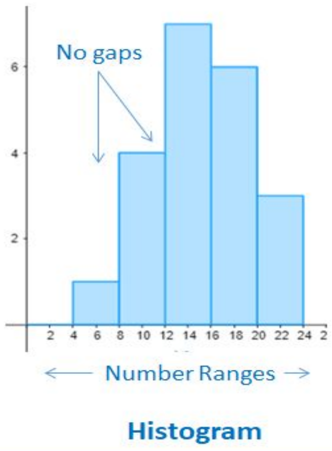
Week 2 Lecture Review

Visualizing Distributions

Bar Charts (categorical variables):
- Y-axis: a count/percentage of each category



Histograms (quantitative variables):
- Distribution of values across bins
- Describe shape, center, spread, outliers



LAB 02 Walkthrough

Lab Submission

- Follow the directions on the LAB02 file
- Submit using the **Terminal Tab** (next to the console in the bottom left pane)
    - Copy and paste the given line into the terminal
    - Follow prompts (NOTE: the terminal will **not** show your password being typed out!)
- **<u>CHECK IN GRADESCOPE THAT ALL YOUR TESTS PASSED</u>**